

Data Wrangling and Data Analysis

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Erik-Jan van Kesteren

Ayoub Bagheri



Utrecht University

Today

- Who are we?
 - Introduction to the course content
 - Getting to know you a little
 - Break
 - Practicalities & course flow
 - Time for questions
-
- But first...



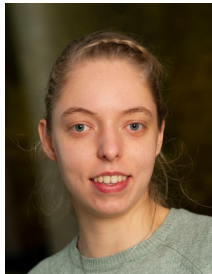
Who are we?



Utrecht University

Faculty of Science team

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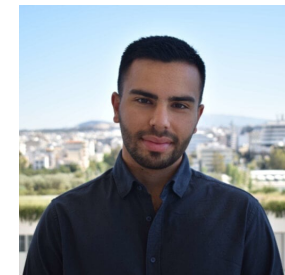
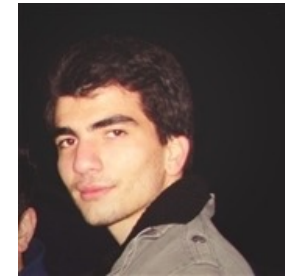
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Lab Group 1

Lab Group 2

Lab Group 3

Social & Behavioural Science team

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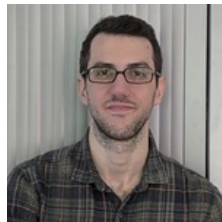
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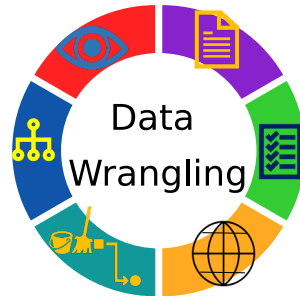
Course introduction



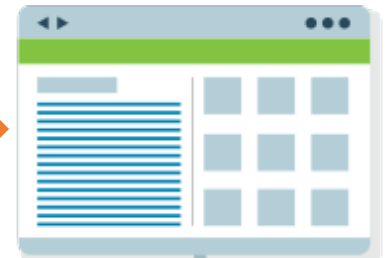
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Extracting value from data

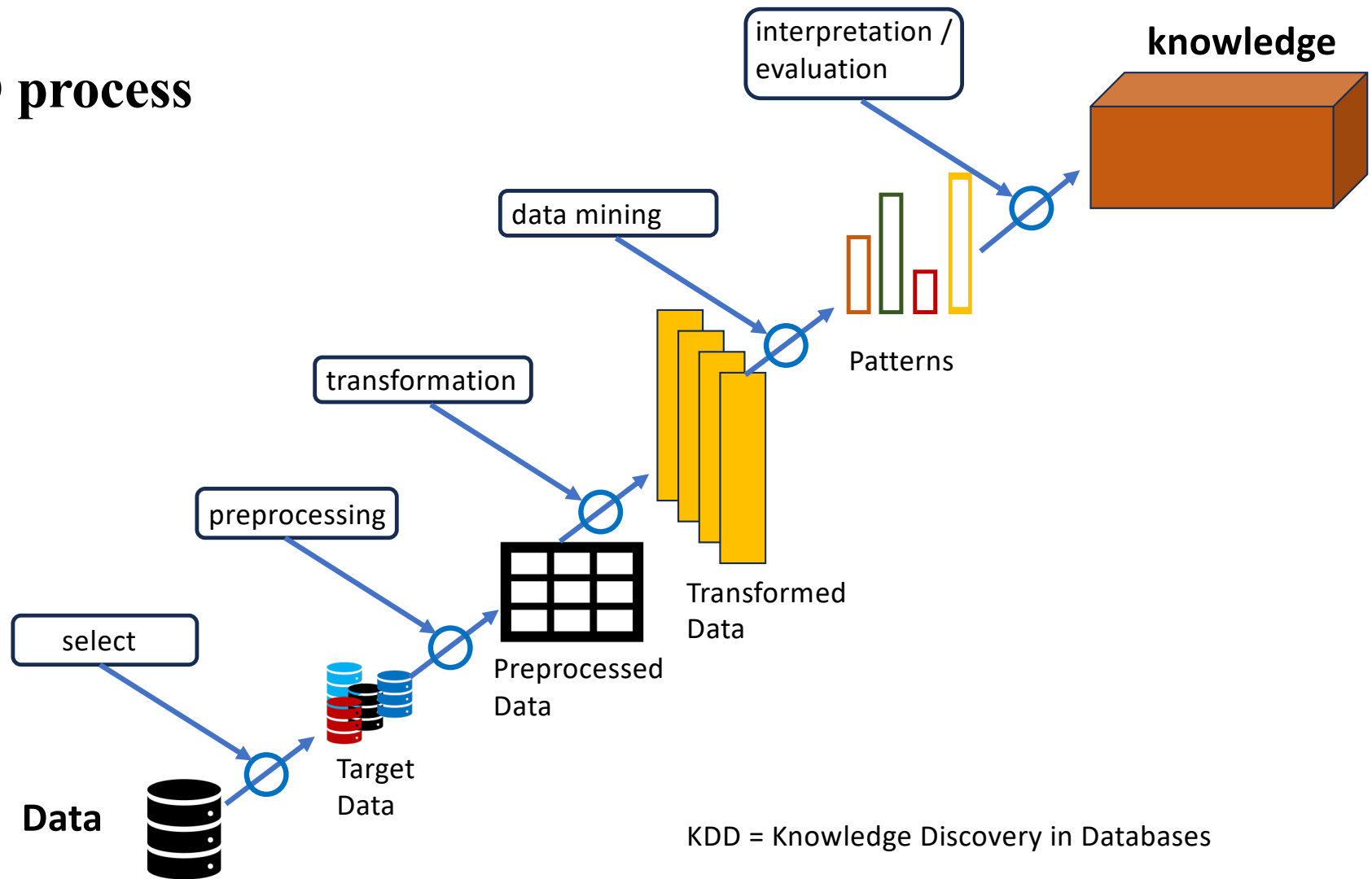
Data



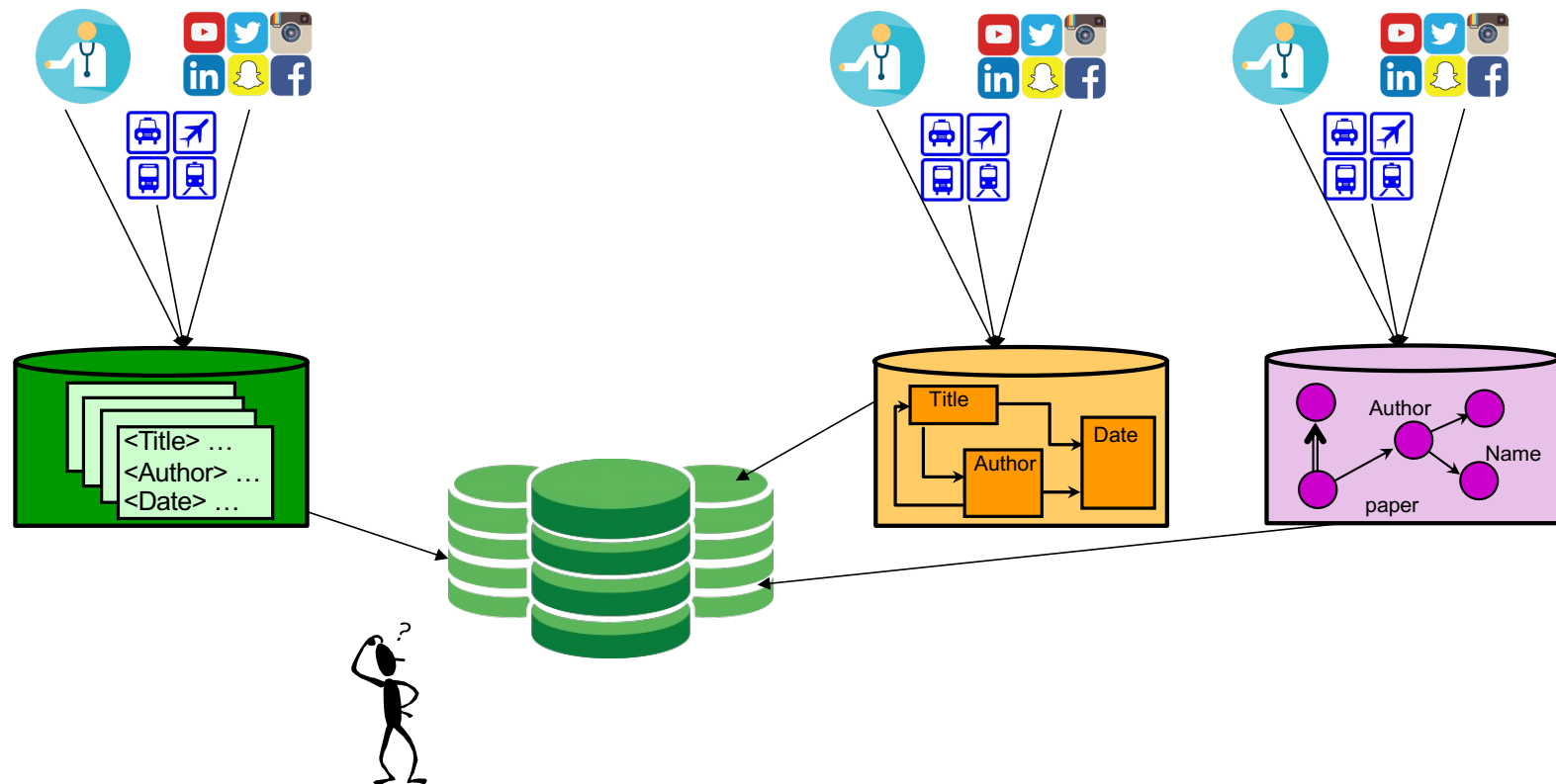
Information



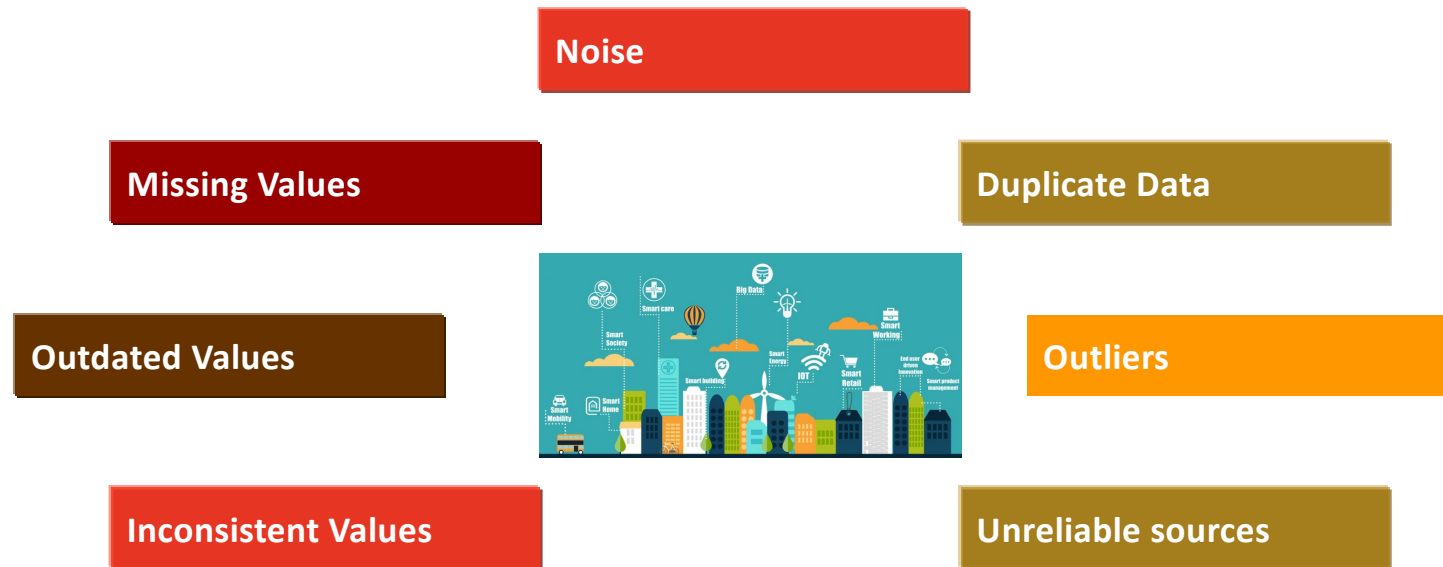
KDD process



We collect a lot of data



Data quality issues

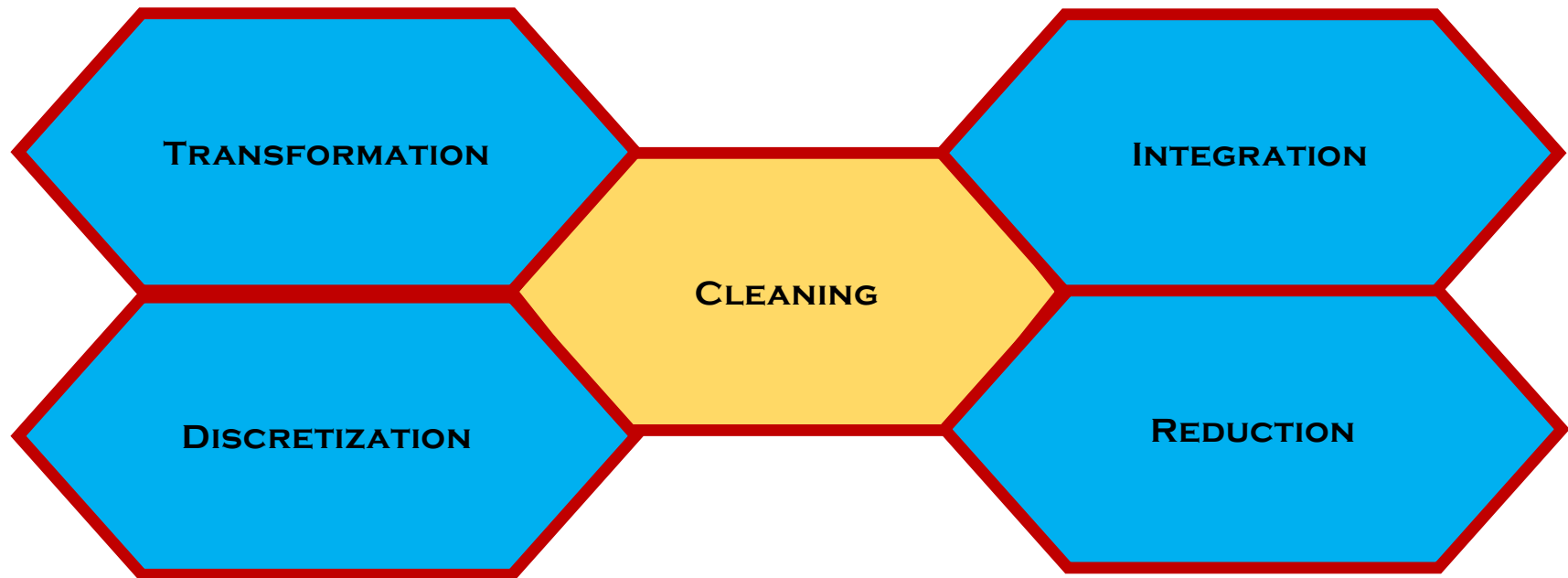


For any data retrieval or data analytics task,
it is critically important to know the quality of your data

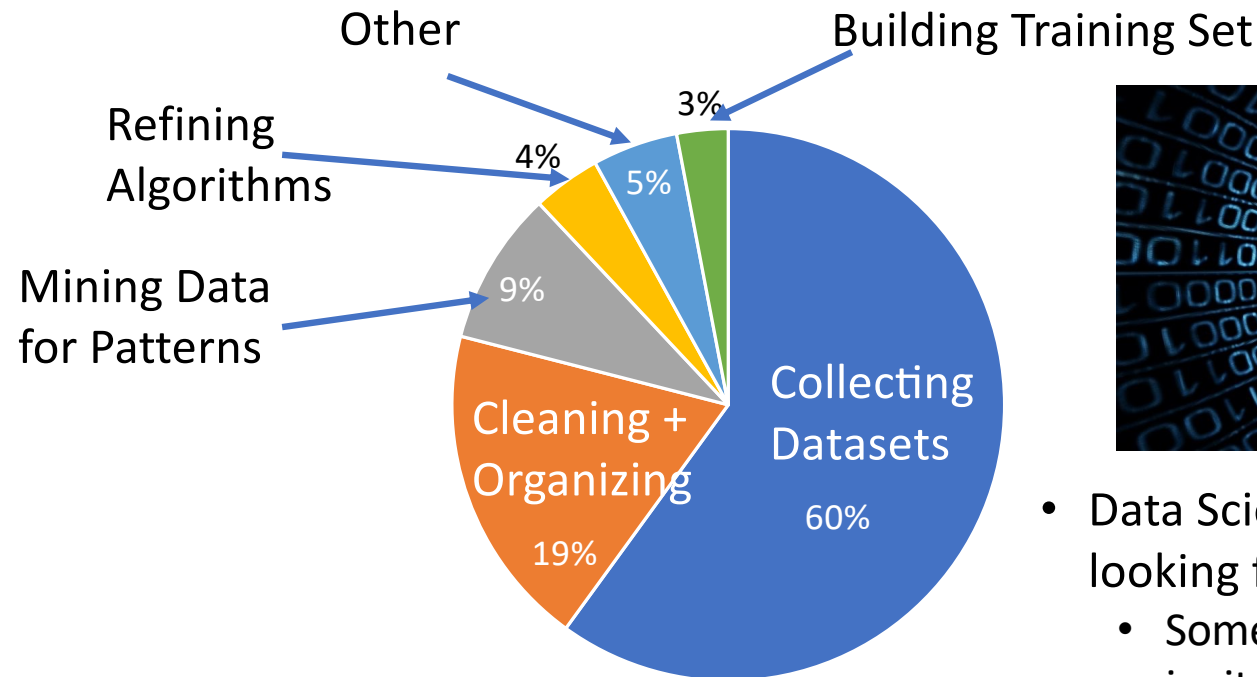
For the US alone, poor data quality costs around US economy \$3 trillions a year (2016).



Data preparation



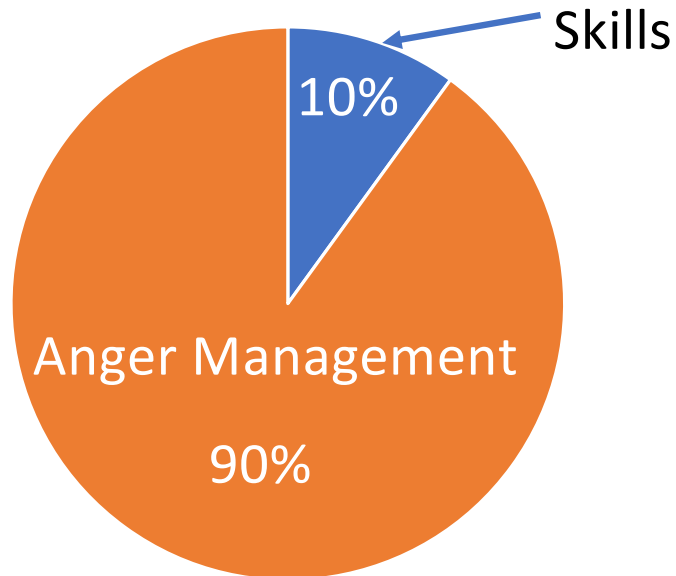
How are you going to spend your time?



- Data Scientists: 80% of their time looking for and cleaning the data
 - Sometimes, they are called data janitors



Data Wrangling



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Regression: How do wages differ?

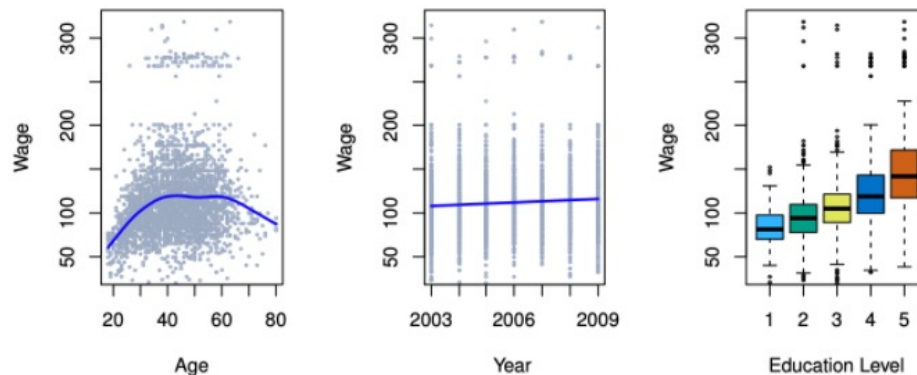
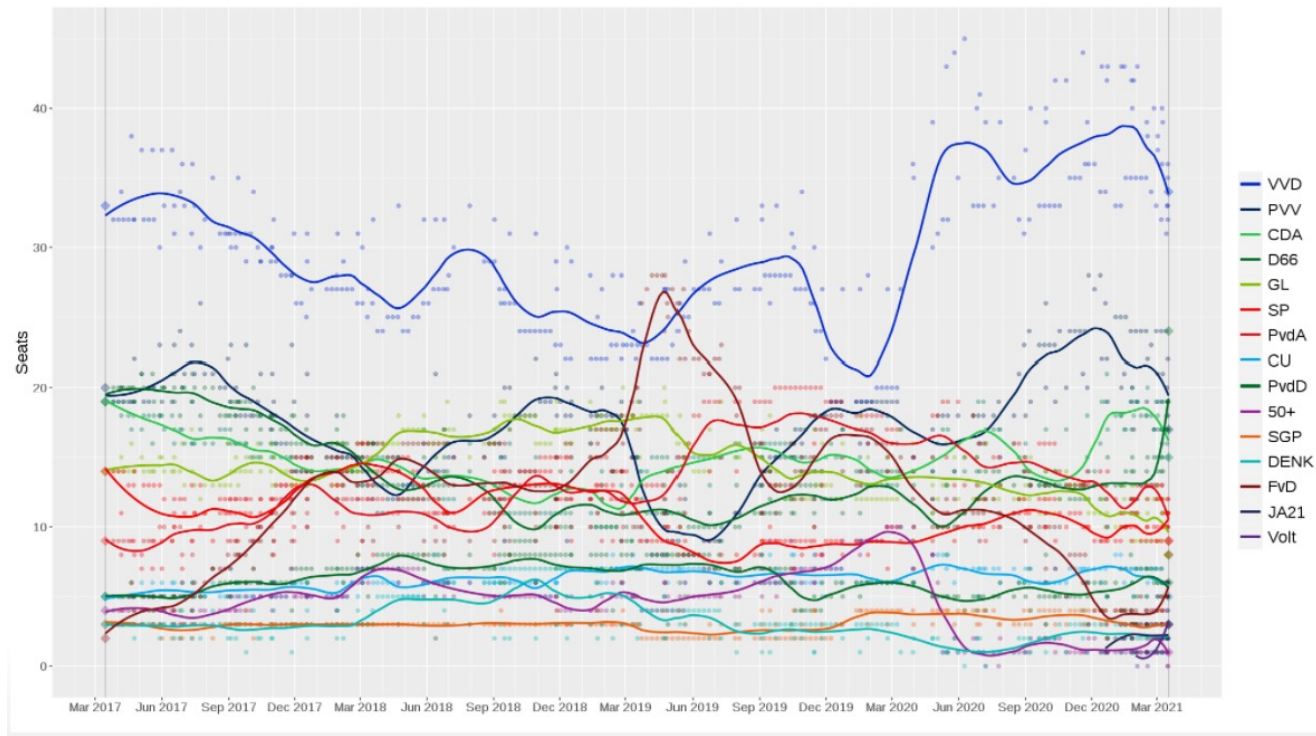


FIGURE 1.1. Wage data, which contains income survey information for men from the central Atlantic region of the United States. Left: wage as a function of age. On average, wage increases with age until about 60 years of age, at which point it begins to decline. Center: wage as a function of year. There is a slow but steady increase of approximately \$10,000 in the average wage between 2003 and 2009. Right: Boxplots displaying wage as a function of education, with 1 indicating the lowest level (no high school diploma) and 5 the highest level (an advanced graduate degree). On average, wage increases with the level of education.



Classification & visualisation



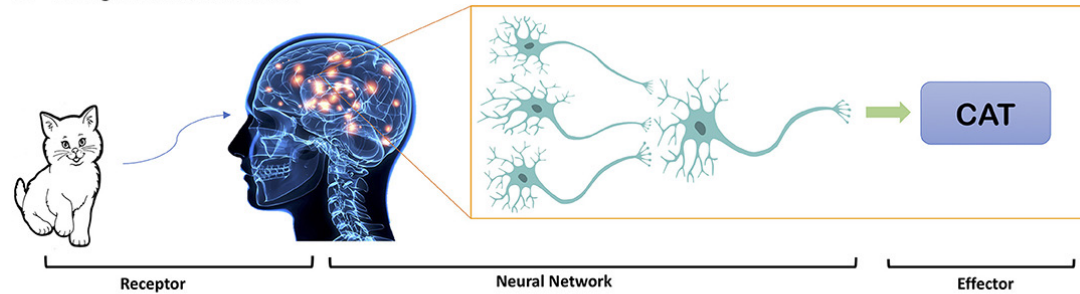
Predicting the outcome of the 2021 Dutch election:
<https://gitlab.com/gbuvn1/opinion-polling-graph>



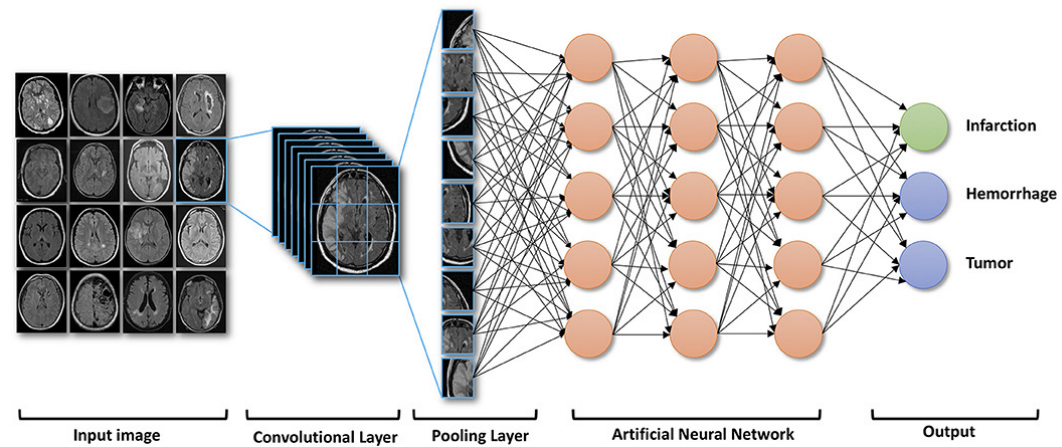
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Deep learning

A Biological Neural Network



B Computer Neural Network(Convolutional Neural Network)



Missing data

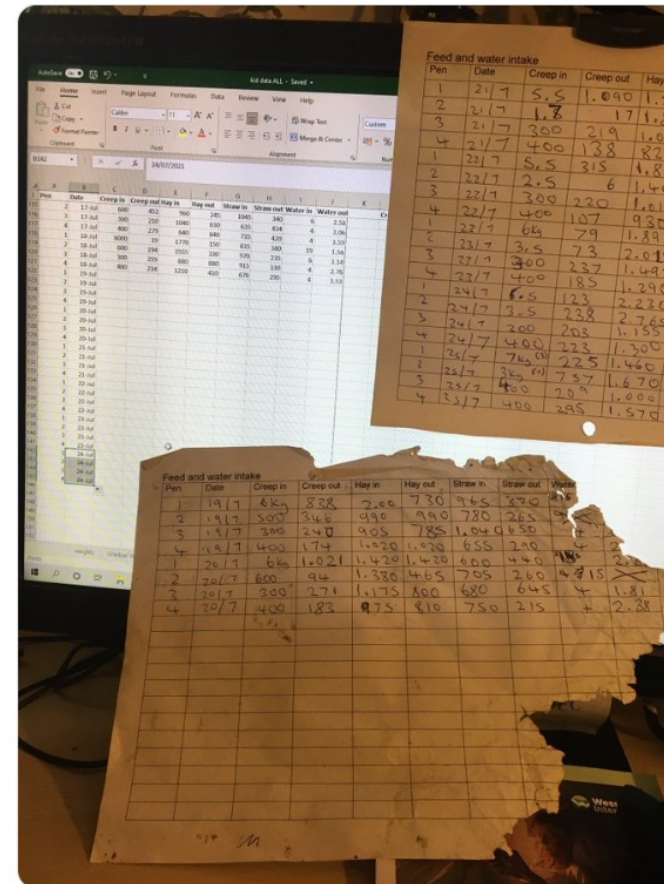
- Missing data mechanisms
- Imputation methods



Holly Vickery PhD

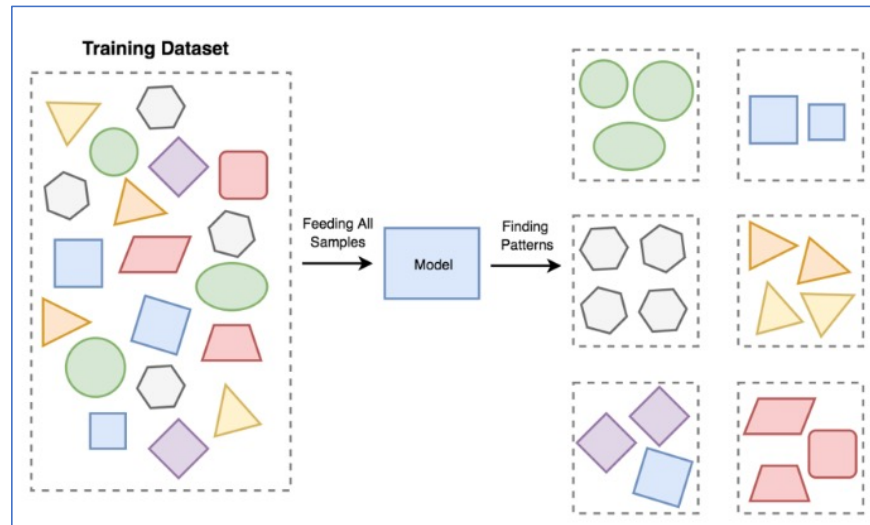
@SkylarkHolly

Working with goats 🐐👩



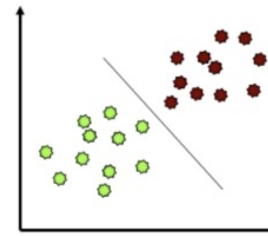
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Clustering



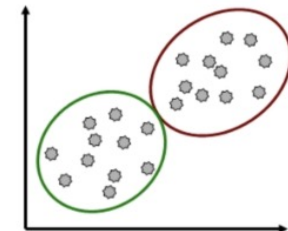
CLASSIFICATION

- Labeled data points
- Want a "rule" that assigns labels to new points
- Supervised learning



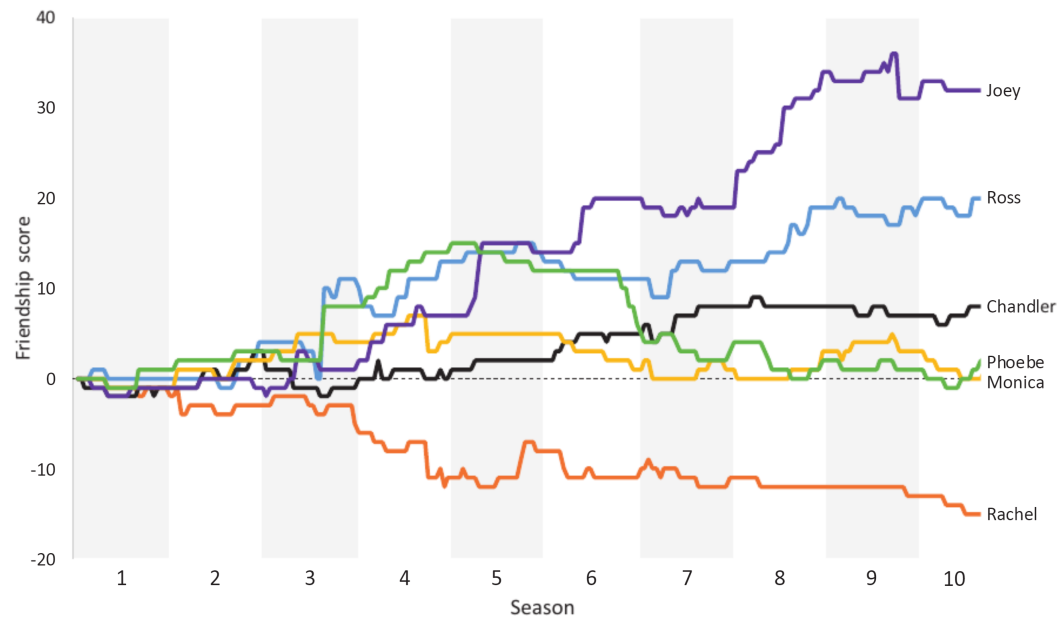
CLUSTERING

- Data is not labeled
- Group points that are "close" to each other
- Identify structure or patterns in data
- Unsupervised learning



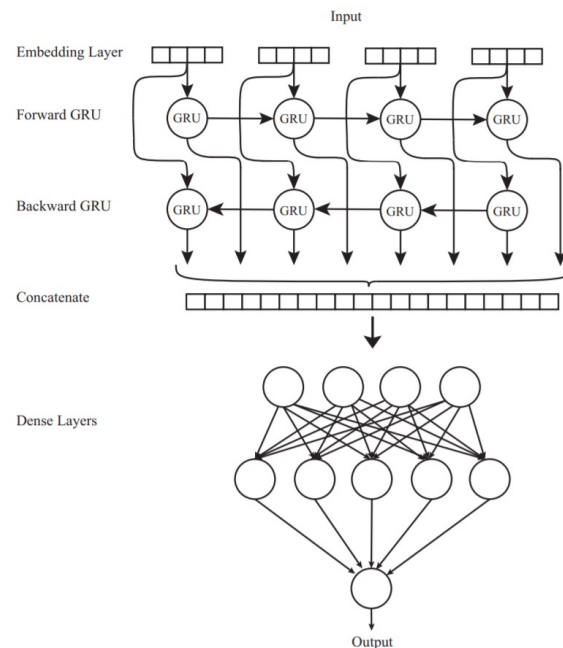
Text Mining: Who was the best Friend?

<https://rss.onlinelibrary.wiley.com/doi/epdf/10.1111/1740-9713.01574>



Natural language processing

<https://www.nature.com/articles/s41746-021-00404-9>



Box 1: An example of a Dutch discharge letter from the dataset

Bovengenoemde patiënt was opgenomen op <DATUM-1> op de <PERSOON-1> voor het specialisme Cardiologie.

Reden van opname STEMI inferior

Cardiale voorgeschiedenis. Blanco

Cardiovasculaire risicofactoren: Roken(-) Diabetes(-) Hypertensie(?) Hypercholesterolemie (?)

Anamnese. Om 18.30 pijn op de borst met uitstraling naar de linkerarm, zweten, misselijk. Ambulance gebeld en bij aansluiten monitor beeld van acuut onderwandinfarct.

AMBU overdracht. 500 mg aspegic iv, ticagrelor 180 mg oraal, heparine, zofran eenmalig, 3x NTG spray. HD stabiel gebleven. Medicatie bij presentatie. Geen.

Lichamelijk onderzoek. Grauw, vegetatief, Halsvenen niet gestuwd. Cor s1 s2 geen souffles. Pulm schoon. Extr warm en slank.

Aanvullend onderzoek. AMBU ECG: Sinusritme, STEMI inferior III/II C/vermoedelijk RCA.

Coronair angiografie. (...). Conclusie angio: 1-vatslijden..PCI

Conclusie en beleid

Bovengenoemde <LEEF TIJD-1>jarige man, blanco cardiale voorgeschiedenis, werd gepresenteerd vanwege een STEMI inferior waarvoor een spoed PCI werd verricht van de mid-RCA. Er bestaan geen relevante nevenletsels. Hij kon na de procedure worden overgeplaatst naar de CCU van het <INSTELLING-2>...Dank voor de snelle overname...Medicatie bij overplaatsing. Acetylsalicylzuur dispersietablet 80 mg; oraal; 1x per dag 80 milligram; <DATUM-1>. Ticagrelor tablet 90 mg; oraal; 2x per dag 90 milligram; <DATUM-1>. Metoprolol tablet 50 mg; oraal; 2x per dag 25 milligram; <DATUM-1>. Atorvastatine tablet 40 mg (als ca-zout-3-water); oraal; 1x per dag 40 milligram; <DATUM-1>

Samenvatting

Hoofddiagnose: STEMI inferior vv PCI RCA. Geen nevenletsels. Nevend diagnoses: geen.

Complicaties: geen Ontslag naar: CCU <INSTELLING-2>.



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NLP: ChatGPT4o



Quiz me on
world capitals



Message to
comfort a friend



Activities to make
friends in new city

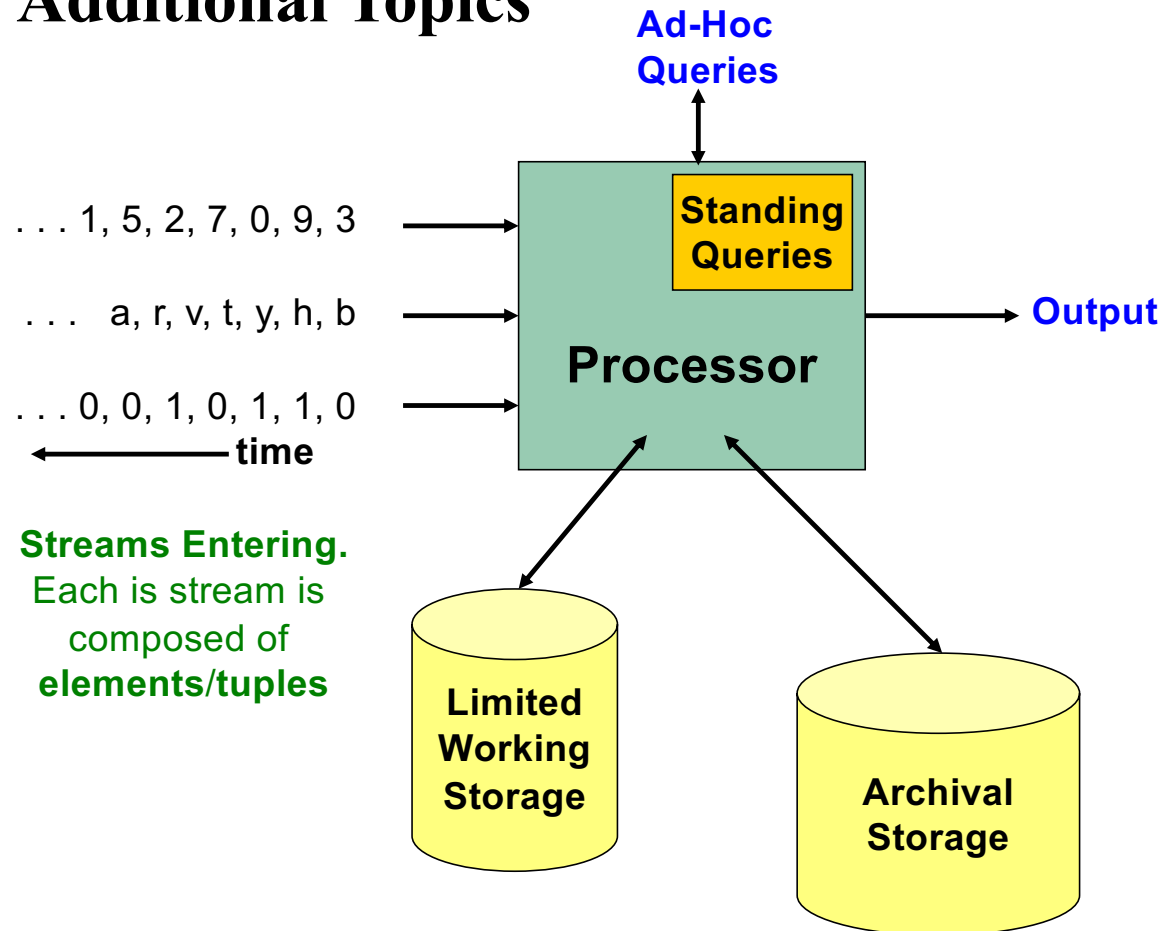


Pick outfit to look
good on camera



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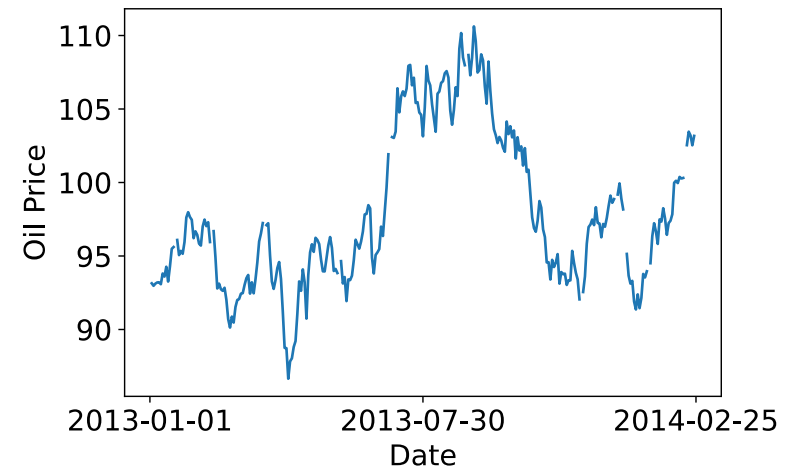
Additional Topics



Streams Entering.
Each is stream is
composed of
elements/tuples



Algorithmic Fairness



This course is useful for your future!

Head of Data Science at [redacted]

APR 23

9:46 AM

Dear Professor Qahtan,

I hope you're doing well.

I was a student in your Data Wrangling and Data Analysis course during the 2021/2022 academic year. Your lecture slides were incredibly helpful. I've often relied on them in my work since then.

Unfortunately, I only had them saved locally, and my computer recently suffered a motherboard failure. Would it be possible for you to share the slides with me again?

I'd really appreciate it. Thanks in advance!

Thank you for Data Wrangling

😊 ↩ ⏪ ⏩

[redacted] @students.u...

Friday, 28 April 2023 at 11:35

With our school year coming to a close, and us having had all our courses. I wanted to take the moment to appreciate you for the effort you have put into data wrangling. Eventhough the course is quite challenging for people without a data science background, like myself, and I did feel less energy the first half of the next semester after the high workload, data wrangling has been the course where I got to learn the most and has been the most interesting for me.



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1

Go to wooclap.com

2

Enter the event code in the top banner

Event code

INFOMDWR

app.wooclap.com/INFOMDWR



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Break



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Practicalities



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<https://infomdwr.nl>



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Practicalities

- Everything is on our course website infomdwr.nl, except:
 - Room locations (on mytimetable.uu.nl)
 - Announcements (in your email and on uu.brightspace.com)
 - Assignment hand-in (on uu.brightspace.com)
 - Exams (on our digital assessment tool Remindo)
 - Lectures / Labs will be in person

 **materials on the website are constantly under construction** 



Course flow

- This is a 14EC course. That's a lot!
- Every* week you will:



Read the required readings



Attend three lectures: Tuesday, Thursday, Friday



Attend and work on three lab sessions: Tuesday, Thursday, Friday



Work on bi-weekly group assignments



Review materials diligently for the midterm and final exams

* some weeks contain other stuff like exams and study time



Schedule & readings

- The syllabus contains a full schedule with required readings

<https://infomdwr.nl/syllabus.html#required-readings>

- Reading materials can be found online for free! 🕵️ (you are resourceful students)



Lectures: Prepare for tomorrow!

1. Look up reading materials in syllabus
2. Look up time and location for the lecture



Labs

- Computer labs with practical exercises
- Put into practice what you learnt in the lectures
- We incorporate real-world data and use-cases: APPLIED data science!
- 3 different sessions for different groups
- **Lab teachers are your main point of contact for questions!**
- Labs provide skills needed to do the assignments



Labs: prepare!

1. Quickly check what today's and tomorrow's labs are about
2. Look up time and location for the labs: mytimetable.uu.nl
3. Which parallel lab group are you assigned to? This should also be reflected in Brightspace



Assignments

- Every two weeks
- Hand in on Brightspace
- Group work!
 - You can assign yourself in a group starting from Friday (tomorrow) until Wednesday next week.
- **Plan ahead**



Assignments: prepare!

1. When is the deadline of the first assignment? infomdwr.nl
2. Quickly check the content of the first assignment
3. Log in to Brightspace and find the INFOMDWR course
4. If we were organized coordinators:
 1. Find the group sign-up page there to see how you can sign up
 2. Find the assignment hand-in page to see how you can submit
5. If we are disorganized coordinators, don't worry, it'll be there soon 😊





Read the syllabus

Announcements



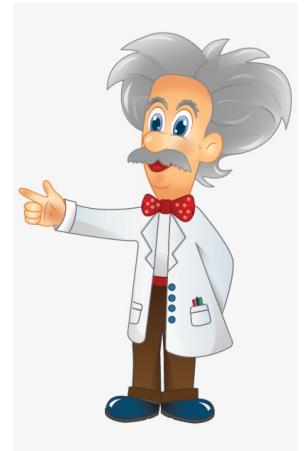
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**Join the Teams channel for communication with the
ADS admin**



ADS OAC is looking for Student Members

- ADS Education Advisory Committee (OAC) is looking for three student members
 - One member (vice-chair) will be a member of GSNS OC
- OAC is composed of equal number of staff and student members.



What will you do?

- You will:
 - Review the Caracal evaluations
 - Comment and give suggestions on the GSNS OER (Education and Examination Rules) and program-wise (ADS) OER Annex
 - Help in solving other education quality-related issues
- Expected time dedication
 - Weekly average: 2 hours for OAC members, 4 hours for vice-chair (also member of GSNS-OC)
 - You will be compensated for the time you spent



How to apply?

- Interested?



+



Angelos Chatzimpampas
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ADS SOCIAL EVENT

LOCATION: The Playground

DATE: TUESDAY SEPTEMBER 16

TIME: 13:00 - 17:00

Come by for a drink and a piece of
Pizza, chat with your peers,
instructors and play some board
games

Volunteers are needed



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Further questions?



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See you in the lab at BBG 201 🖐️
First lab is optional!

